

SAFETY DATA SHEET

FORMULA 410

1. PRODUCT AND COMPANY IDENTIFICATION

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CHEMTREC 24 HOUR EMERGENCY RESPONSE (800) 424-9300

Product Number	410
Product Name	Water Sciences Formula 410
Chemical Family	Water Treatment Compound, Liquid
Date Prepared	12/2015
Revision Date	05/2016
Recommended Use	Corrosion Inhibitor
Restrictions	Not intended for use where the treated steam may contact dairy products.

2. HAZARDOUS IDENTIFICATION

GHS CLASSIFICATION

Classification

Flammable Liquids	Category 4
Corrosive to Metals	Category 1
Acute Toxicity, Oral	Category 4
Acute Toxicity, Dermal	Category 4
Skin Corrosion/Irritation	Category 2
Sensitization, Skin	Category 1
Serious Eye Damage/Eye Irritation	Category 2A
Acute Toxicity, Inhalation	Category 4
Specific Target Organ Toxicity, Single Exposure, R	Category 3
Reproductive Toxicity	Category 2
Hazardous to the Aquatic Environment, Long-Term Hazard	Category 3

WARNING!

GHS LABEL:



2. HAZARDOUS IDENTIFICATION - continued

HAZARD STATEMENTS

H227	Combustible liquid
H290	May be corrosive to metals
H302	Harmful if swallowed
H312	Harmful in contact with skin
H315	Causes skin irritation
H317	May cause an allergic skin reaction
H319	Causes serious eye irritation
H332	Harmful if inhaled
H335	May cause respiratory irritation
H361	Suspected of damaging fertility or the unborn child
H412	Harmful to aquatic life with long lasting effects

PRECAUTIONARY STATEMENTS

P201	Causes skin irritation
P202	Do not handle until all safety precautions have been read and understood
P210	Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. NO SMOKING
P234	Keep only in original packaging
P261	Avoid breathing dust/fume/gas/mist/vapor/spray
P264	Wash thoroughly after handling
P270	Do not eat, drink, or smoke when using this product
P271	Use only outdoors or in a well-ventilated area
P272	Contaminated work clothing should not be allowed out of the workplace
P273	Avoid release to the environment
P280	Wear protective gloves/protective clothing/eye protection/face protection
P301+P312	IF SWALLOWED: Call a POISON CONTROL CENTER for doctor/physician if you feel unwell
P302+P352	IF ON SKIN: Wash with plenty of water.
P304+P340	IF INHALED: Remove victim to fresh air and keep comfortable for breathing
P305+P351-P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308+P313	If exposed or concerned, get medical advice/attention
P312	Call a POISON CONTROL CENTER or a doctor/physician if you feel unwell
P321	Specific treatment (see label)
P330	Rinse mouth
P332+P313	If skin irritation occurs, get medical attention/advice
P333+P313	If skin irritation or rash occurs, get medical attention/advice
P337+P313	If eye irritation persists, get medical attention/advice
P362+P364	Take off immediately all contaminated clothing and wash it before re-use
P370+P378	In case of fire: Use water spray to extinguish
P390	Absorb spillage to prevent material damage
P403+P233+P405	Store in a well-ventilated area. Keep container tightly closed. Store locked up.
P406	Store in a corrosive resistant container with a resistant inner liner.
P501	Dispose of contents/container to hazardous or special waste collection point

3. COMPOSITION/INFORMATION ON INGREDIENTS

The precise composition of this product is proprietary information. In the event of a medical emergency, a complete disclosure will be provided to medical personnel.

Component Name	CAS #	Component %	OSHA PEL	ACGIH TLV
Cyclohexylamine	108-91-8	8-10%	10 ppm	10 ppm
Morpholine	110-91-8	8-10%	20 ppm	20 ppm
Diethylhydroxylamine	3710-84-7	5-8%	None Listed	None Listed

4. FIRST AID MEASURES

CONTACT WITH EYES	Immediately wash affected eyes for at least 15 minutes under running water with eyelids help open. Consult an eye specialist.
SKIN CONTACT	Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.
INHALATION	Keep patient calm. Remove to fresh air and seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.
INGESTION	Immediately rinse mouth and then drink 200-300 mL of water. Seek medical attention.

5. FIREFIGHTING MEASURES

SUITABLE EXTINGUISHING MEDIA	Water spray, fog, or mist. Alcohol resistant foam. DO NOT use CO ₂ or dry chemicals.
SPECIAL FIREFIGHTING PROCEDURES	Use self-contained breathing apparatus and full bunker gear in fire areas. Evacuate all unprotected personnel from area. Keep containers cool with water fog to minimize swelling, taking care not to spread flames with water used for cooling. Do not flush down sewers or other drainage systems.
UNUSUAL FIREFIGHTING HAZARDS	Combustion products: Thermal decomposition products may include oxides of carbon and nitrogen.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS	Keep unnecessary people away; isolate hazard area and deny entry. Stay upwind; keep out of low areas. (Also see section 8). Wear suitable protective equipment. Do not touch damaged containers or spilled material unless wearing suitable protective clothing.
ENVIRONMENTAL PRECAUTIONS	Contain spill if it can be done with minimal risk. Prevent liquid from entering drains, sewers or waterways. Notify proper authorities. Take up with sand or other noncombustible absorbent material and place into containers for later disposal.
METHODS FOR CLEANING UP	SMALL SPILLS: Take up with sand or other noncombustible absorbent material and place into containers for later disposal. LARGE SPILLS: Dike far ahead of liquid spill for later disposal.

7. HANDLING AND STORAGE

Store materials to avoid sources of ignition and excessive heat.

Do not get in eyes, on skin, or on clothing. Do not breath vapors or mists. Keep container closed. Use only with adequate ventilation. Use good personal hygiene practices. Wash hands before eating, drinking, or smoking.

Remove contaminated clothing and clean before re-use.

8. PRECAUTIONS TO CONTROL EXPOSURE/PERSONAL PROTECTION

EXPOSURE LIMITS:

Component Name	CAS #	OSHA PEL	ACGIH TLV
Diethylhydroxylamine	3710-84-7	None Listed	None Listed
Morpholine	110-91-8	20 ppm	20 ppm
Cyclohexylamine	108-91-8	10 ppm	10 ppm

Engineering Controls: Adequate local or mechanical to reduce vapor or mist to below the PEL or TLV. Local exhaust ventilation may be necessary to control air contaminants to their exposure limits. The use of local ventilation is recommended to control emissions near the source. Provide mechanical ventilation for confined spaces. Use explosion-proof ventilation equipment.

Monitoring: Do not eat, drink, or smoke in areas where this chemical is used or stored. Wash hands prior to eating, drinking, or using the restroom.

Personal Protective Equipment (PPE):

- **Eye Protection:** Wear chemical safety goggles. Have eye-wash stations available where eye contact can occur.
- **Skin Protection:** Avoid skin contact. Wear gloves impervious to conditions of use. Additional protection may be necessary to prevent skin contact, including use of apron, face shield, boots or full body protection. A safety shower should be located in the work area.
- **Respiratory Protection:** If exposure limits are exceeded, NIOSH approved respiratory protection should be worn. A NIOSH approved respirator for organic vapors is generally acceptable for concentrations up to 10 times the PEL. For higher concentrations, unknown concentrations and for oxygen deficient atmospheres, use a NIOSH approved air-supplied respirator. Engineering controls are the preferred means for controlling chemical exposures. Respiratory protection may be needed for non-routine or emergency situations. Respiratory protection must be provided in accordance with OSHA 29 CFR 1910.134.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE	Clear to Pale Yellow Liquid
ODOR	Fishy
pH @ 25° C	11.0 – 12.25
MELTING/FREEZING POINT	Not Known
FLASHPOINT	>140° F
SPECIFIC GRAVITY	0.96
SOLUBILITY	Complete
AUTO-IGNITION TEMPERATURE	Not Known
DECOMPOSITION TEMPERATURE	Not Known
VOC CONTENT	% Volatile by Volume: 100
ODOR THRESHOLD	Not Known
BOILING RANGE	100 - 110° C
EVAPORATION POINT	Not Known
FLAMMABLE LIMITS – UPPER	Not Known
FLAMMABLE LIMITS – LOWER	Not Known
VAPOR PRESSURE	7 mmHg
VAPOR DENSITY (AIR=1)	Not Known
VISCOSITY	Water-like

10. STABILITY AND REACTIVITY

Stability	Stable under normal conditions of storage and handling
Conditions to Avoid	Avoid all sources of ignition: heat, sparks, and open flame. Avoid electro-static charge.
Hazardous Decomposition/By Products	Possible thermal decomposition products: carbon oxides, nitrogen oxides
Hazardous Polymerization	Will not occur
Polymerization Conditions to Avoid	None
Incompatibilities	Strong acids and oxidizers

11. TOXICOLOGICAL INFORMATION

Likely route of Exposure	Routes of entry for solids and liquids are ingestion and inhalation but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquified gases.
Inhalation	Breathing spray or mists can be harmful. Can cause damage to nasal and respiratory passages.
Eye Contact	Causes eye irritation including stinging, watering and redness, which may result in corneal injury.
Skin Contact	Contact may cause mild stinging irritation including redness, burning and drying/cracking of the skin.

11. TOXICOLOGICAL INFORMATION - continued
Ingestion

Aspiration hazard. Can enter lungs during swallowing or vomiting and cause chemical pneumonia and edema.

Acute Toxicity Value

See Health Hazards Below.

Assessment of acute toxicity: Moderate toxicity after single ingestion. Pronounced toxicity after short-term skin contact. Virtually nontoxic by inhalation. The inhalation of highly enriched or saturated vapor-air-mixture represents an acute hazard.

Chronic (Long-Term) Effects

See Health Hazards Below. Genetic toxicity Assessment of mutagenicity: Most of the results from the available studies show no evidence of a mutagenic effect. Carcinogenicity Assessment of carcinogenicity: In long-term studies in rats and mice in which the substance was given by feed, a carcinogenic effect was not observed. Literature data. Reproductive toxicity Assessment of reproduction toxicity: On the basis of animal study findings show no effect on fertility. Literature data. Teratogenicity Assessment of teratogenicity: The substance did not cause malformations in animal studies. When given in high doses embryo toxicity was observed. Literature data.

Component Name	LD50	LC50
Diethylhydroxylamine	Dermal Rabbit – 1,300 mg/kg Oral Rat – 2,190 mg/kg	Inh. Rat – 11.4 mg/l
Morpholine	Dermal Rabbit – 500 mg/kg Oral Rat – 1,1910 mg/kg	Inh Rat – 8 mg/l
Cyclohexylamine	Oral Rat – 300 mg/kg Dermal Rabbit – 277 mg/kg	Inh. Rat – 7,500 mg/m3

Reproductive Effects

Fertility

Teratogenicity

Not Applicable

Mutagenicity

Not Applicable

Embryotoxicity

Not Applicable

Sensitization to Product

Skin

Synergistic Products

Not Applicable

Carcinogenicity

Not Applicable

12. ECOLOGICAL INFORMATION
Ecotoxicity

Harmful to aquatic life. Toxicity to fish LC50 (96h) 237 mg/l, Oncorhynchus mykiss (static). The details of the toxic effect relate to the nominal concentration. The study was carried out in soft water. Literature data. Aquatic invertebrates EC50 (48h) 110 mg/l, Daphnia magna (DIN 38412 Par 11, static). The details of the toxic effect relate to the nominal concentration.

Mobility

Information not available

12. ECOLOGICAL INFORMATION

Degradability Expected to biodegrade

Bioaccumulation Information not available

13. WASTE DISPOSAL CONSIDERATIONS

Waste Disposal of Substance: Dispose of in accordance with national, state and local regulations.

If containers are not empty, they must be disposed of in a RCRA-licensed facility. RCRA empty containers may be landfilled at a licensed facility; other containers must be disposed of in a RCRA licensed facility.

RCRA: D001

14. TRANSPORT INFORMATION

DOT SHIPPING INFORMATION

UN: 2735

Proper Shipping Name Amines, Liquid, Corrosive, N.O.S.

Contains: Cyclohexylamine, Morpholine

Hazard Class and Label 8

Packaging Group II

Other Shipping Info: **SPECIAL PRECAUTIONS FOR USER:** Always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code: Not

15. REGULATORY INFORMATION

TSCA STATUS: The components of this product are listed on the TSCA Inventory.

SARA TITLE III SECTION 302/304 EXTREMELY HAZARDOUS SUBSTANCE:

Component Name	CAS #	% By Weight	RQ (Lbs.)	TPQ (Lbs.)
Cyclohexylamine	108-91-8	10%	10,000	10,000

SARA TITLE III SECTION 311/312 HAZARD CATEGORIZATION:

Acute	Chronic	Fire	Pressure	Reactive
X	X	N/A	N/A	N/A

SARA TITLE III SECTION 313 SUPPLIER INFORMATION:

No chemicals in this material are subject to the reporting requirements.

CERCLA SECTION 102(a) HAZARDOUS SUBSTANCE:

Component Name	CAS #	% By Weight	RQ (Lbs.)
Morpholine	110-91-8	10%	100 Lbs.

CALIFORNIA PROPOSITION 65:

This product contains the following chemical(s) known to the State of California to cause cancer, birth defects, or other reproductive harm. CAS# 110-91-8 (110-91-8)

16. OTHER INFORMATION

HMIS Health	3
HMIS Flammability	2
HMIS Reactivity	0

Additional: Federal Law. Also contact appropriate State and Local regulatory agencies.
49CFR 172.101, Appendix A, Table 1, Reportable Quantity: No components are listed.
Toxic Substances Control Act (TSCA): All components of this product are included on the TSCA inventory.
Clean Water Act (CWA): Not Applicable
Component CWA-Hazardous Substances CWA Reportable Quantities:
CWA-Toxic Pollutants
CWA-Priority Pollutants: Cyclohexylamine 1,000 Lbs.

Prepared by: Mike Fuze, Water Sciences Disclaimer: The information contained herein is accurate to the best of our knowledge. Water Sciences makes no warranty of any kind, express or implied, concerning the safe use of this material in your process or in combination with other substances.